Batch size for curvature steps: three quantities

Derivations, background, and why they might matter

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Question. At which batch size does an exact (layerwise) GGN step stop overfitting its batch, while keeping the advantage the quadratic model promises? This note derives three quantities that answer it: a noise-to-signal ratio r that decides both overfitting and how much of the advantage survives (Section 2); a compute-optimal batch size once steps have a price (Section 3); and the noise in the curvature estimate itself, which classical batch-size theory leaves out (Section 4). Section 5 combines them, and Section 6 lists what to investigate. Statements are exact unless labelled as approximations; all numbers marked as checks come from `make_figure.py` in this folder.

1 Setting

Near the current weights w, model the corpus loss by a quadratic,

$$L(w) = L^* + \tfrac{1}{2}(w - w^*)^\top H (w - w^*), \qquad g = H(w - w^*), \qquad \Delta L = L(w) - L^* = \tfrac{1}{2}\, g^\top H^{-1} g.$$

A batch of B examples drawn without replacement from N gives $g_B = g + \xi$ with $\mathbb{E}\xi = 0$ and $\operatorname{Cov}\xi = (\frac{1}{B} - \frac{1}{N})\Sigma$, where Σ is the covariance of per-example gradients. A preconditioned step is $w^+ = w - \alpha P g_B$ with P symmetric positive definite. The exact Newton (GGN) step is $P = H^{-1}$; for least squares the GGN equals H. For Newton it helps to whiten: with $u = H^{1/2}(w - w^*)$ the loss is $\Delta L = \frac{1}{2}\|u\|^2$, the whitened gradient is u, and the whitened noise is $\zeta = H^{-1/2}\xi$.

2 Quantity 1: the ratio r, overfitting, and what survives

Proposition 1 (Step size tuned in expectation)**.** *For fixed P, the best expected decrease of the corpus loss is*

$$\max_\alpha \mathbb{E}\big[L(w) - L(w^+)\big] = \frac{\Phi_P}{1 + r_P}, \qquad \Phi_P = \frac{(g^\top P g)^2}{2\, g^\top PHPg},$$

$$r_P = \Big(\frac{1}{B} - \frac{1}{N}\Big) K_P, \qquad K_P = \frac{\operatorname{tr}(PHP\Sigma)}{g^\top PHPg}.$$

Proof. $L(w) - L(w^+) = \alpha g^\top P g_B - \frac{1}{2}\alpha^2 g_B^\top PHP g_B$. Taking expectations with $\mathbb{E}g_B = g$ and $\mathbb{E}[g_B^\top M g_B] = g^\top M g + \operatorname{tr}(M \operatorname{Cov}\xi)$ gives $\alpha\, g^\top P g - \frac{1}{2}\alpha^2 (g^\top PHPg + \operatorname{tr}(PHP \operatorname{Cov}\xi))$; maximize over α. □

Φ_P is what the quadratic model promises for the direction without noise; K_P is the step's *noise scale*, in examples. For Newton, $\Phi = \Delta L$ (the noiseless Newton step solves the quadratic), and

$$\text{share kept} = \frac{1}{1 + r}, \qquad r = \Big(\frac{1}{B} - \frac{1}{N}\Big) K_1, \qquad K_1 = \frac{\operatorname{tr}(H^{-1}\Sigma)}{2\,\Delta L}.$$

For least squares with label noise σ^2, near the optimum $\Sigma \approx \sigma^2 H$ and therefore $K_1 \approx D\sigma^2/(2\Delta L)$: the noise scale grows as the loss falls.

Approximation 2 (Step size fitted on the step's own batch)**.** Let $\hat{\alpha}$ minimize the batch's own quadratic model along Pg_B (with the exact curvature H). The share of the decrease the batch model predicts that the corpus actually sees is exactly $s = 2\,g^\top Pg_B/(g_B^\top Pg_B) - 1$. Replacing numerator and denominator by their expectations,

$$s \approx \frac{1-\tilde{r}}{1+\tilde{r}}, \qquad \tilde{r} = \frac{\operatorname{tr}(P \operatorname{Cov} \xi)}{g^\top Pg},$$

and $\tilde{r} = r$ for Newton. Check (whitened Newton, $D = 200$): $r = \frac{1}{2}$ gives 0.336 against 0.333; $r = 2$ gives -0.332 against -0.333.

Derivation of the exact form. The fitted step size is $\hat{\alpha} = g_B^\top Pg_B/(g_B^\top PHPg_B)$, and the batch-predicted decrease is $\frac{1}{2}\hat{\alpha}\, g_B^\top Pg_B$. The corpus decrease is

$$\hat{\alpha}\, g^\top Pg_B - \tfrac{1}{2}\hat{\alpha}^2 g_B^\top PHPg_B = \hat{\alpha}\, g^\top Pg_B - \tfrac{1}{2}\hat{\alpha}\, g_B^\top Pg_B.$$

Divide the second by the first. □

Reading. One ratio decides both questions in the opening sentence.

- *Overfitting:* for $r > 1$, i.e. $B \lesssim K_1$, a Newton step whose length is fitted on its own batch *increases* the corpus loss in expectation.
- *Advantage kept:* with a step size chosen in expectation (or on a second batch), a fraction $1/(1+r)$ of the promised gain survives.

Background. For $P = I$ this is McCandlish et al.'s noise scale: the best progress per step is $\Delta L_{\max}/(1 + B_{\text{noise}}/B)$, and the trade-off between steps and examples has its knee near B_{noise}. Zhang et al.'s noisy quadratic model showed that preconditioning raises the batch size up to which larger batches help. The share law of Approximation 2 is the lab's transfer law (N16/N20 in `ideas/batch_noise.md`): on MNIST it predicted measured shares with Spearman 0.963. It overstated the harm for the exact GGN at small B, where the batch's curvature along the fitted step exceeds the corpus curvature, so the search limits itself.

Why it matters here. In `noise_scale_20261021` (CIFAR, whitened per-tensor GGN), r at $B = 2{,}048$ was 0.63–1.01 at step 10 and 8.1–9.5 at step 400. So in theory about half of the Newton advantage survives early and about a tenth late; the measured gains were smaller still. This is the lab's rule "curvature pays early, not late" as a formula. Because K_P grows with the strength of P, it also explains why the best power of the curvature is fractional.

3 Quantity 2: the compute-optimal batch

The share $1/(1+r)$ improves with B indefinitely, so "the right batch size" needs a price. Let one step cost $c(B) = c_{\text{step}} + c_{\text{ex}}B$: a fixed part per step (building and factorizing curvature, solves, overheads) and a part per example. Write $B_{\text{ov}} = c_{\text{step}}/c_{\text{ex}}$, the per-step overhead in units of examples.

Proposition 3 (Compute-optimal batch)**.** *For $N \to \infty$ and fixed P, the expected progress per unit compute, $\pi(B) = \Phi_P/((1 + K_P/B)(c_{\text{step}} + c_{\text{ex}}B))$, is maximized at*

$$B^* = \sqrt{K_P\, B_{\text{ov}}}, \qquad \textit{where the share kept is } \frac{1}{1+\sqrt{K_P/B_{\text{ov}}}}.$$

Proof. $\pi \propto 1/h(B)$ with $h(B) = (1 + K/B)(c_{\text{step}} + c_{\text{ex}}B) = c_{\text{step}} + c_{\text{ex}}K + c_{\text{ex}}B + Kc_{\text{step}}/B$. Then $h'(B) = c_{\text{ex}} - Kc_{\text{step}}/B^2 = 0$ at $B^2 = Kc_{\text{step}}/c_{\text{ex}}$, and h is convex for $B > 0$. □

Remark 4.

(a) A step with almost no per-step overhead ($B_{\text{ov}} \approx 0$, plain SGD) wants small batches; a step that pays a lot per step wants larger ones. The optimum is the geometric mean of the noise scale and the overhead.

(b) For a dense Newton step on a D-dimensional block, $c_{\text{ex}} \approx D^2$ (accumulating the curvature) and $c_{\text{step}} \approx D^3/3$ (Cholesky), so $B_{\text{ov}} \approx D/3$. For least squares this gives $B^* \approx D\sigma/\sqrt{6\Delta L}$: the compute-optimal batch grows as the excess loss falls, i.e. the theory asks for a growing batch schedule.

(c) Lab costs are measurable: a full-GGN product in `bulk_krylov_20261014` took 0.070 s at 256 examples and 0.46 s at 2,048. A line through these two points gives about 14 ms fixed plus 2.2×10^{-4} s per example, so the fixed part equals about 66 examples (two data points; an illustration only).

Background and why it matters. This is a two-line consequence of McCandlish et al.'s trade-off with an explicit price per step; it is elementary and probably stated somewhere. Its use here is that it turns "what does curvature cost" (the Hennig route) into a batch-size rule, and the growth of K_1 during training (the Orvieto route, where the noise scale acts as a temperature) into a batch schedule. It is also the natural target of the derivative-free homework: a compute-limited run whose optimum the theory predicts.

4 Quantity 3: noise in the curvature itself

A GGN optimizer does not know H; it estimates it from a batch. Classical batch-size theory, including both references above, treats the curvature as known. Take least squares with Gaussian inputs $x \sim \mathcal{N}(0, H)$ and curvature $\hat{H} = \frac{1}{n}\sum_{i \le n} x_i x_i^\top$ from n examples. In whitened coordinates $\hat{S} = H^{-1/2}\hat{H}H^{-1/2}$, and $n\hat{S}$ is a Wishart matrix with n degrees of freedom and identity scale.

Lemma 5 (Inverse-Wishart moments; e.g. Gupta & Nagar 2000)**.** *For $n > D + 3$,*

$$\mathbb{E}[\hat{S}^{-1}] = m_1 I, \quad m_1 = \frac{n}{n - D - 1}; \quad \mathbb{E}[\hat{S}^{-2}] = m_2 I, \quad m_2 = \frac{n^2(n-1)}{(n-D)(n-D-1)(n-D-3)}.$$

Check ($D = 20$, $n = 60$, 20,000 draws): m_1 1.5382 against 1.5385, m_2 3.6770 against 3.6798.

So the batch Newton step is too long on average, by $n/(n - D - 1)$, and its spread blows up as n approaches D: the smallest eigenvalues of a sample covariance are underestimated, and the inverse amplifies them.

Proposition 6 (Both noise channels)**.** *Take the Newton direction with curvature $\hat{S}$ from n examples and a gradient from an independent batch with ratio r (Section 2). With the best step size, the expected fraction of ΔL removed is*

$$\frac{\kappa(n, D)}{1 + r}, \qquad \kappa(n, D) = \frac{m_1^2}{m_2} = \frac{(n-D)(n-D-3)}{(n-D-1)(n-1)}.$$

Check ($D = 50$, $n = 120$, $r = 1$): 0.2853 simulated against 0.2856.

Proof. In whitened coordinates the step is $u^+ = u - \alpha\hat{S}^{-1}(u + \zeta)$, with ζ independent of $\hat{S}$, $\mathbb{E}\zeta = 0$ and $\mathbb{E}\|\zeta\|^2 = r\|u\|^2$. By Lemma 5 and independence, $\mathbb{E}\|u^+\|^2 = \|u\|^2(1 - 2\alpha m_1 + \alpha^2 m_2(1 + r))$. The minimum over α is $\|u\|^2(1 - m_1^2/(m_2(1 + r)))$. □

Remark 7.

(a) $\kappa \approx (n - D - 2)/(n - 1) \approx 1 - D/n$ for $n \gg D$: curvature noise acts like a second noise scale of about D examples, and the two channels multiply. As n falls to $D + 3$, $\kappa \to 0$: below about D examples per block, an undamped Newton step gains nothing even with the best step size.

(b) *GGN heuristic.* One example contributes rank at most $C-1$ to a GGN with C outputs, so the threshold is roughly $n \approx D/(C-1)$. For the lab's 64×256 MLP blocks ($D = 16{,}384$, $C = 10$) that is about 1,820 examples, close to the training batch of 2,048. This is a heuristic: GGN rows are not Gaussian, confident predictions shrink the effective rank, and damping changes the picture near the threshold.

(c) *Fractional powers.* The mean of $s^{-\nu}$ over the sample spectrum increases with ν, so curvature noise penalizes $\nu = 1$ more than $\nu = \frac{1}{2}$. That is the direction of the noise-scale study's miss: its theory treated the curvature as exact and overstated the gains by 1.5–3.8× at $\nu = \frac{1}{2}$ and 2–9× at $\nu = 1$.

Background and why it matters. The underestimated small eigenvalues are the Marchenko–Pastur picture of sample covariances. In optimization, Tatzel et al. (2025) show that mini-batch quadratics are systematically biased, the same phenomenon seen from the quadratic model's side. Of the three quantities this is the one least covered by standard batch-size theory: it has its own threshold, independent of gradient noise, and it is specific to curvature methods. It is a candidate explanation for the open question of why the noise-scale theory was too optimistic (open test 6 in `ideas/batch_noise.md`).

5 Putting them together

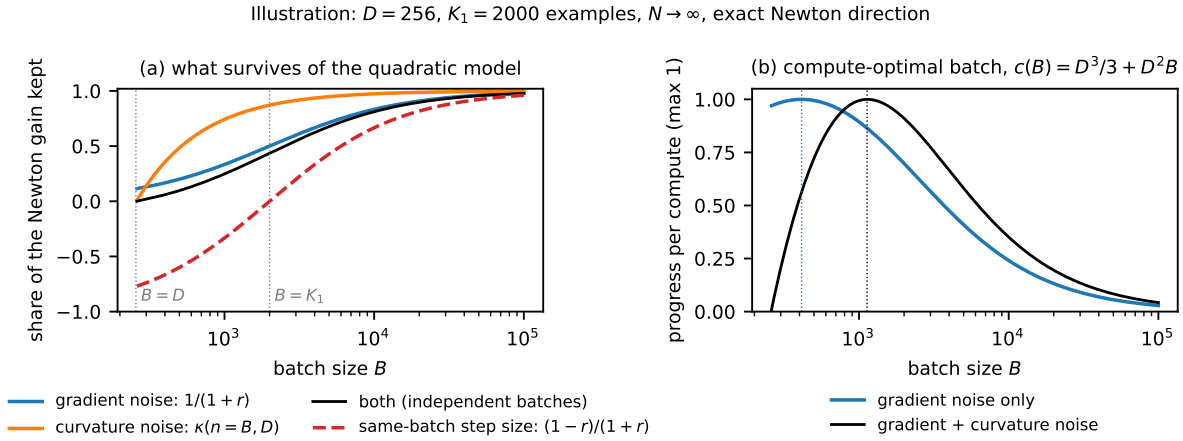


Figure 1: Illustration with $D = 256$, $K_1 = 2{,}000$, $N \to \infty$, exact Newton direction. (a) Share of the Newton gain kept: gradient noise (Proposition 1), curvature noise with $n = B$ (Proposition 6), both from independent batches, and the share with a step size fitted on the batch (Approximation 2). (b) Progress per unit compute with $c(B) = D^3/3 + D^2B$, i.e. $B_{\text{ov}} \approx 85$.

In the illustration of Figure 1, gradient noise alone puts the compute-optimal batch at $B^* = \sqrt{K_1 B_{\text{ov}}} = 413$ (numerical maximum 413). There only 0.17 of the gain survives the gradient noise, and curvature noise from a batch of that size keeps only 0.38 of the rest. With both channels the optimum moves to about 1132, a few multiples of D. Curvature noise thus pushes the compute-optimal batch of a curvature method up, toward the block dimension. The figure assumes independent batches for gradient and curvature; the case of one batch for both is not covered by these formulas.

6 What to investigate

1. **Synthetic least squares.** Verify Proposition 6 with independent batches, then use one batch for gradient and curvature ($n = B$, correlated; the step becomes the batch's own least-squares solution), then add damping δ. How should the best δ scale with D/n?
2. **Compute-optimal batch by derivative-free search** (the homework). Inputs $\log B$, $\log \delta$, $\log \eta$ under a fixed compute budget; compare the optimum found with Proposition 3 and Figure 1.
3. **Curvature channel in the lab** (open test 6). Replay noise-scale states with the curvature estimated from $n \in \{1{,}000,\ 4{,}000,\ 16{,}000\}$ reference examples. Does the theory's optimism shrink as n grows?
4. **Cost in the lab.** Measure c_{step} and c_{ex} for an actual curvature step, predict $B^*(t)$ from the measured $K_1(t)$, and test a growing batch schedule against a constant $B = 2{,}048$ at equal compute.
5. **Averaging.** The lab's optimizers average statistics over steps. An average over k steps lowers the noise roughly k-fold but adds staleness; fold this into all three quantities (T4 in `ideas/batch_noise.md`).

References

[1] A. K. Gupta and D. K. Nagar. *Matrix Variate Distributions*. Chapman & Hall/CRC, 2000. (Cited from memory; the two moments in Lemma 5 were checked numerically.)

[2] S. McCandlish, J. Kaplan, D. Amodei et al. An empirical model of large-batch training. arXiv:1812.06162, 2018. (Cited from memory.)

[3] L. Tatzel, B. Mucsányi, O. Hackel and P. Hennig. Debiasing mini-batch quadratics for applications in deep learning. *ICLR*, 2025. arXiv:2410.14325. (Checked online 23 October 2026.)

[4] G. Zhang, L. Li, Z. Nado, J. Martens, S. Sachdeva, G. Dahl, C. Shallue and R. Grosse. Which algorithmic choices matter at which batch sizes? Insights from a noisy quadratic model. *NeurIPS*, 2019. arXiv:1907.04164. (Cited from memory.)

[5] Lab notes: `ideas/batch_noise.md` (T1, T7, R3, open test 6); `studies/noise_scale_20261021/RESULTS.md`; `studies/bulk_krylov_20261014/RESULTS.md` (K5).